
Appendix A: State Machine Diagrams

A.1 Overview

This appendix consolidates all state machine diagrams used throughout the NewPOPSys platform. These diagrams define the valid states, transitions, and triggers for key business entities.

State Machine Categories

Category	Entities	Owner
Campaign Lifecycle	Campaign	Brand Portal
Store Execution	StoreAssignment, AssignmentItem	Store App / Brand Portal
Fulfillment	StoreOrder, Shipment, FulfillmentStatus	PSP Portal
Verification	PhotoReview, IssueRequest	Brand Portal

A.2 Campaign Lifecycle States

The campaign progresses through a defined lifecycle from creation to archival.

A.2.1 Campaign Status State Diagram

```
stateDiagram-v2
    [*] --> DRAFT : Create Campaign

    DRAFT --> SCHEDULED : Set Future Start Date
    DRAFT --> PUBLISHED : Publish Now
    DRAFT --> CANCELLED : Cancel Campaign

    SCHEDULED --> PUBLISHED : Start Time Reached
    SCHEDULED --> DRAFT : Unschedule
    SCHEDULED --> CANCELLED : Cancel Campaign

    PUBLISHED --> COMPLETED : All Stores Satisfied OR End Date Reached
    PUBLISHED --> CANCELLED : Terminate Early

    COMPLETED --> ARCHIVED : Archive Campaign
    CANCELLED --> ARCHIVED : Archive Campaign

    ARCHIVED --> [*]
```

A.2.2 Campaign State Definitions

State	Description	Allowed Actions
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DRAFT	Campaign being configured; not visible to stores	Edit all fields, delete, schedule, publish
SCHEDULED	Set to publish at a future date	Edit until publish time, unschedule, cancel
PUBLISHED	Active campaign; stores can execute	Add stores, monitor, terminate
COMPLETED	All stores satisfied or end date passed	View reports, archive
CANCELLED	Terminated before natural completion	View history, archive
ARCHIVED	Historical record; read-only	View only

A.2.3 Campaign Transition Triggers

From	To	Trigger	Actor
DRAFT	SCHEDULED	<code>campaign.schedule(startDate)</code>	Brand Admin
DRAFT	PUBLISHED	<code>campaign.publish()</code>	Brand Admin
SCHEDULED	PUBLISHED	System scheduler (cron)	System
PUBLISHED	COMPLETED	Rollup satisfies threshold OR <code>endDate <= now</code>	System
PUBLISHED	CANCELLED	<code>campaign.terminate(reason)</code>	Brand Admin
*	ARCHIVED	<code>campaign.archive()</code>	Brand Admin

A.3 Store Assignment States

Tracks each store's progress through a campaign from assignment to completion.

A.3.1 StoreAssignment Status State Diagram

```
stateDiagram-v2
    [*] --> ASSIGNED : Campaign Published

    ASSIGNED --> READY : Materials Delivered OR Manual Allow
    READY --> IN_PROGRESS : Store Starts Work
    IN_PROGRESS --> SUBMITTED : Proofs Submitted

    SUBMITTED --> COMPLETE : All Photos Approved
    SUBMITTED --> WAIVED : Admin Waives Compliance
    SUBMITTED --> REWORK_REQUIRED : Photos Rejected

    REWORK_REQUIRED --> IN_PROGRESS : Retake Started
    REWORK_REQUIRED --> SUBMITTED : Retake Submitted

    COMPLETE --> REOPENED : Admin Reopen
    REOPENED --> IN_PROGRESS : Work Continues
```

WAIVED --> [*]
COMPLETE --> [*]

A.3.2 StoreAssignment State Definitions

State	Description	Store Can...	Admin Can...
ASSIGNED	Store selected for campaign	View campaign details	Edit assignment
READY	Materials delivered; execution allowed	Begin installation	Force ready status
IN_PROGRESS	Store actively working	Capture photos, report issues	Monitor progress
SUBMITTED	All photos submitted for review	Wait for review	Review photos
COMPLETE	All requirements satisfied	View completion	Reopen if needed
REWORK_REQUIRED	Photos rejected; needs retry	View feedback, retake	Send reminders
REOPENED	Admin reopened for additional work	Continue execution	Close again
WAIVED	Admin bypassed verification	N/A	View waiver reason

A.4 Assignment Item States

Tracks individual kit items at a store through the fulfillment and execution lifecycle.

A.4.1 AssignmentItem Status State Diagram

```
stateDiagram-v2
    [*] --> PLANNED : Assignment Created

    PLANNED --> ORDERED : Order Line Created
    ORDERED --> SHIPPED : Shipment Line Created
    SHIPPED --> DELIVERED : Carrier Delivered
    DELIVERED --> RECEIVED : Store Verified Receipt

    RECEIVED --> INSTALLED : Installation Recorded
    RECEIVED --> EXCEPTION : Issue Reported (Missing/Damaged)

    INSTALLED --> PROOF_SUBMITTED : Photo Uploaded
    PROOF_SUBMITTED --> SATISFIED : Photo Approved
    PROOF_SUBMITTED --> RETAKE_REQUIRED : Photo Rejected

    RETAKE_REQUIRED --> PROOF_SUBMITTED : Retake Resubmitted
```

SATISFIED --> [*]
EXCEPTION --> [*]

A.4.2 AssignmentItem State Definitions

State	Description	Next Expected Action
PLANNED	Item assigned to store, not yet ordered	PSP creates order
ORDERED	Order line exists in PSP system	PSP ships item
SHIPPED	Item in transit to store	Carrier delivers
DELIVERED	Carrier confirmed delivery	Store verifies receipt
RECEIVED	Store verified item received	Store installs
INSTALLED	Installation recorded	Store uploads photo
PROOF_SUBMITTED	Photo pending review	Admin reviews
SATISFIED	Photo approved; item complete	None
RETAKE_REQUIRED	Photo rejected; needs retry	Store retakes photo
EXCEPTION	Issue reported (escalated)	Issue resolution flow

A.5 Fulfillment States

Tracks order and shipment progress from PSP perspective.

A.5.1 FulfillmentStatus State Diagram (Computed)

```
stateDiagram-v2
    [*] --> NOT_SHIPPED : Order Created

    NOT_SHIPPED --> PARTIALLY_SHIPPED : shippedQty > 0
    PARTIALLY_SHIPPED --> SHIPPED : shippedQty >= requiredQty

    SHIPPED --> PARTIALLY_DELIVERED : deliveredQty > 0
    PARTIALLY_DELIVERED --> DELIVERED : deliveredQty >= requiredQty

    NOT_SHIPPED --> EXCEPTION : Carrier Exception
    PARTIALLY_SHIPPED --> EXCEPTION : Carrier Exception
    SHIPPED --> EXCEPTION : Carrier Exception
    PARTIALLY_DELIVERED --> EXCEPTION : Carrier Exception

    EXCEPTION --> NOT_SHIPPED : Resolved (No Shipments)
    EXCEPTION --> PARTIALLY_SHIPPED : Resolved (Some Shipped)
    EXCEPTION --> SHIPPED : Resolved (Fully Shipped)

    NOT_SHIPPED --> CANCELLED : Order Cancelled
    PARTIALLY_SHIPPED --> CANCELLED : Order Cancelled
```

```
CANCELLED --> [*]
DELIVERED --> [*]
```

A.5.2 FulfillmentStatus Computation Logic

Status	Condition
NOT_SHIPPED	shippedQty = 0
PARTIALLY_SHIPPED	0 < shippedQty < requiredQty
SHIPPED	shippedQty >= requiredQty
PARTIALLY_DELIVERED	0 < deliveredQty < shippedQty
DELIVERED	deliveredQty >= requiredQty
EXCEPTION	Carrier reports exception flag
CANCELLED	Order marked cancelled

A.6 Photo Review States

Tracks individual photo review workflow.

A.6.1 PhotoReview Status State Diagram

```
stateDiagram-v2
    [*] --> PENDING : Photo Uploaded

    PENDING --> APPROVED : Reviewer Approves
    PENDING --> REJECTED : Reviewer Rejects

    REJECTED --> SUPERSEDED : Retake Photo Uploaded
    SUPERSEDED --> PENDING : New Photo Enters Review

    APPROVED --> [*]
```

A.6.2 PhotoReview State Definitions

State	Description	Next Action
PENDING	Photo submitted, awaiting review	Reviewer approves/rejects
APPROVED	Photo meets requirements	None (terminal)
REJECTED	Photo fails requirements	Store uploads retake
SUPERSEDED	Original replaced by retake	New photo reviewed

A.6.3 Rejection Reason Codes

Code	Description
BLURRY	Image not in focus
WRONG_ANGLE	Incorrect camera angle
OBSTRUCTED	Item partially blocked
WRONG_ITEM	Incorrect item photographed
NOT_INSTALLED	Item not properly installed
POOR_LIGHTING	Insufficient lighting
OTHER	Free-text reason required

A.7 Issue Request States

Tracks issue reporting, triage, and resolution workflow.

A.7.1 IssueRequest Status State Diagram

```
stateDiagram-v2
    [*] --> OPEN : Store Reports Issue

    OPEN --> TRIAGED : Classified and Routed
    TRIAGED --> AWAITING_APPROVAL : Needs Brand Approval
    TRIAGED --> APPROVED : Auto-Approved by Policy
    TRIAGED --> DENIED : Rejected

    AWAITING_APPROVAL --> APPROVED : Brand Approved
    AWAITING_APPROVAL --> DENIED : Brand Denied

    APPROVED --> IN_FULFILLMENT : Reorder Created
    IN_FULFILLMENT --> RESOLVED : Replacement Shipped and Received

    DENIED --> [*]
    RESOLVED --> [*]
```

A.7.2 IssueRequest State Definitions

State	Description	Responsible Party
OPEN	Issue reported by store	System (auto-triage)
TRIAGED	Classified and routed	System / PSP Admin
AWAITING_APPROVAL	Requires brand approval	Brand Admin
APPROVED	Approved for reorder	PSP Operations
DENIED	Issue rejected	None (closed)

IN_FULFILLMENT	Replacement being shipped	PSP Operations
RESOLVED	Replacement received	None (closed)

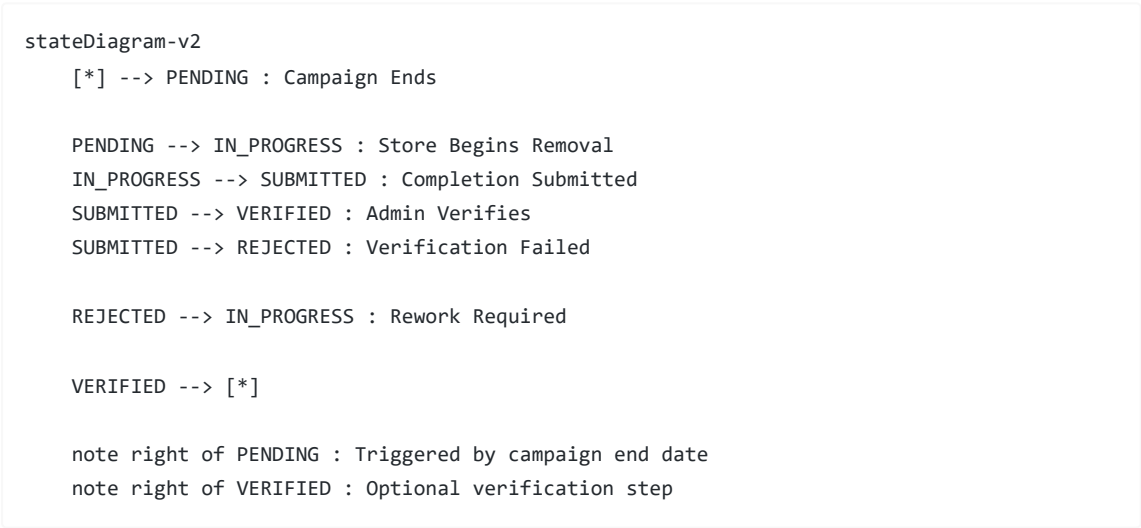
A.7.3 Issue Type Classifications

Type	Description	Auto-Approve Eligible
MISSING	Item not in shipment	Yes (if < threshold)
DAMAGED	Item arrived damaged	Yes (with photo)
INCORRECT	Wrong item shipped	No (requires review)
PACKAGING	Packaging issue only	No

A.8 Deinstall Task States

Tracks end-of-campaign material removal workflow.

A.8.1 DeinstallTask Status State Diagram



A.8.2 DeinstallTask State Definitions

State	Description
PENDING	Task created when campaign ends
IN_PROGRESS	Store actively removing materials
SUBMITTED	Store marked removal complete
VERIFIED	Admin confirmed removal (optional)
REJECTED	Removal not complete; rework needed

A.9 Store Order States

Tracks PSP order processing workflow.

A.9.1 StoreOrder Status State Diagram

```
stateDiagram-v2
    [*] --> PENDING : Order Created

    PENDING --> CONFIRMED : PSP Acknowledges
    CONFIRMED --> IN_PRODUCTION : Entered Production
    IN_PRODUCTION --> READY_TO_SHIP : Production Complete
    READY_TO_SHIP --> SHIPPED : All Lines Shipped
    SHIPPED --> DELIVERED : Carrier Delivered

    PENDING --> CANCELLED : Campaign Cancelled
    CONFIRMED --> CANCELLED : Order Cancelled
    IN_PRODUCTION --> ON_HOLD : Production Issue
    ON_HOLD --> IN_PRODUCTION : Issue Resolved

    DELIVERED --> [*]
    CANCELLED --> [*]
```

A.9.2 StoreOrder State Definitions

State	PSP Action	Store Visibility
PENDING	Review order details	"Order Placed"
CONFIRMED	Acknowledge receipt	"Order Confirmed"
IN_PRODUCTION	Produce materials	"In Production"
READY_TO_SHIP	Pick, pack, ship	"Ready to Ship"
SHIPPED	Tracking provided	"Shipped" with tracking
DELIVERED	Carrier confirmed	"Delivered"
ON_HOLD	Resolve issue	"Delayed"
CANCELLED	Order cancelled	"Cancelled"

A.10 State Interrelationships

A.10.1 Status Ownership by Module

```
flowchart LR
    subgraph PSP_STATUS["PSP-Owned"]
        OS["StoreOrderStatus"]
        SS["ShipmentStatus"]
    end
```



```

end

subgraph STORE_STATUS["Store-Owned"]
    RS["ReceiptStatus"]
    ES["ExecutionStatus"]
end

subgraph BRAND_STATUS["Brand-Owned"]
    VS["VerificationStatus"]
    PRS["PhotoReviewStatus"]
end

subgraph DERIVED["Derived (Computed)"]
    FS["FulfillmentStatus"]
    SP["StorePhase"]
end

OS --> FS
SS --> FS
FS --> SP
RS --> SP
ES --> SP
VS --> SP
PRS --> VS

```

A.10.2 StorePhase Computation

The `StorePhase` is a computed rollup representing overall store progress:

Phase	Criteria
PENDING	Assigned but materials not shipped
SHIPPING	Materials shipped, not delivered
RECEIVING	Delivered, receipt not complete
EXECUTING	Receipt complete, installation in progress
VERIFYING	Photos submitted, awaiting review
COMPLETE	All items satisfied or waived
EXCEPTION	Has unresolved issues

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Appendix B: Notification Matrix

B.1 Overview

This appendix defines all system-generated notifications, their recipients, delivery methods, and trigger conditions. Notifications ensure timely communication across all user roles throughout the campaign lifecycle.

Notification Channels

Channel	Description	Use Case
Email	Standard email delivery	Non-urgent notifications, summaries
Push	Mobile/PWA push notification	Time-sensitive store actions
In-App	Notification center / badge	All notifications (persistent)
SMS	Text message (optional)	Critical escalations (premium)
Webhook	API callback to external systems	Integration with third-party systems

Notification Priority Levels

Priority	Description	Delivery
CRITICAL	Requires immediate action	Push + Email + In-App
HIGH	Important, action within 24h	Push + In-App
NORMAL	Standard notification	In-App + Email (digest)
LOW	Informational only	In-App only

B.2 Campaign Lifecycle Notifications

Event	PSP Admin	Brand Admin	Campaign Mgr	Regional Mgr	Store Mgr	Store Staff	Method	Priority
Campaign Created (Draft)	-	In-App	In-App	-	-	-	In-App	LOW
Campaign Published	Email	Email, Push	Email, Push	In-App	Push	Push	Multi	HIGH
Campaign Nearing End (7d)	-	Email	Email	In-App	Push	In-App	Multi	NORMAL
Campaign Ending Today	-	Push	Push	Push	Push	Push	Push	HIGH
Campaign Completed	Email	Email, In-App	Email, In-App	In-App	In-App	-	Multi	NORMAL
Campaign Cancelled	Email	Email, Push	Email, Push	In-App	Push	Push	Multi	HIGH

Campaign Archived	-	In-App	In-App	-	-	-	In-App	LOW
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B.3 Store Assignment Notifications

Event	PSP Admin	Brand Admin	Campaign Mgr	Regional Mgr	Store Mgr	Store Staff	Method	Priority
Store Assigned to Campaign	-	In-App	In-App	In-App	Email, Push	Push	Multi	HIGH
Store Removed from Campaign	-	In-App	In-App	In-App	Email, Push	Push	Multi	NORMAL
Store Ready (Materials Delivered)	-	-	-	-	Push	Push	Push	HIGH
Store Started Installation	-	-	In-App	In-App	-	-	In-App	LOW
Store Submitted for Review	-	In-App	Push	Push	In-App	-	Multi	NORMAL
Store Marked Complete	-	In-App	In-App	In-App	Push	Push	Multi	NORMAL
Store Reopened	-	In-App	In-App	In-App	Push	Push	Multi	HIGH
Store Waived	-	In-App	In-App	In-App	Email	-	Multi	NORMAL

B.4 Fulfillment Notifications

Event	PSP Admin	Brand Admin	Campaign Mgr	Regional Mgr	Store Mgr	Store Staff	Method	Priority
Order Created	In-App	-	-	-	-	-	In-App	NORMAL
Order Confirmed	-	-	-	-	Email	-	Email	LOW

Order Shipped	In-App	-	-	-	Push, Email	Push	Multi	HIGH
Tracking Updated	-	-	-	-	In-App	In-App	In-App	LOW
Order Delivered	In-App	-	-	-	Push	Push	Push	HIGH
Delivery Exception	Push	In-App	In-App	In-App	Push, Email	Push	Multi	CRITICAL
Order Cancelled	In-App	Email	Email	In-App	Push, Email	Push	Multi	HIGH
Partial Shipment Sent	In-App	-	-	-	Push	Push	Multi	NORMAL

B.5 Issue & Reorder Notifications

Event	PSP Admin	Brand Admin	Campaign Mgr	Regional Mgr	Store Mgr	Store Staff	Method	Priority
Issue Reported	Push	In-App	In-App	Push	In-App	In-App	Multi	HIGH
Issue Needs Approval	-	Push, Email	Push, Email	-	-	-	Multi	HIGH
Issue Approved	In-App	In-App	In-App	In-App	Push	Push	Multi	NORMAL
Issue Denied	-	In-App	In-App	In-App	Push, Email	Push	Multi	NORMAL
Reorder Created	Push	In-App	In-App	-	In-App	-	Multi	NORMAL
Replacement Shipped	In-App	-	-	-	Push	Push	Multi	HIGH
Issue Resolved	In-App	In-App	In-App	In-App	Push	Push	Multi	NORMAL

B.6 Photo Review & Verification Notifications

Event	PSP Admin	Brand Admin	Campaign Mgr	Regional Mgr	Store Mgr	Store Staff	Method	Priority
Photo Submitted	-	In-App	In-App	Push	-	-	Multi	NORMAL

Photos Pending Review (Batch)	-	Email	Email	Push, Email	-	-	Multi	HIGH
Photo Approved	-	-	-	-	In-App	In-App	In-App	LOW
Photo Rejected	-	-	-	-	Push	Push	Push	HIGH
Retake Requested	-	In-App	In-App	In-App	Push, Email	Push	Multi	CRITICAL
Retake Submitted	-	In-App	In-App	Push	In-App	-	Multi	NORMAL
All Photos Approved	-	In-App	In-App	In-App	Push	Push	Multi	NORMAL
Review Deadline Approaching	-	Push	Push	Push	-	-	Push	HIGH

B.7 User & Access Notifications

Event	PSP Admin	Brand Admin	Campaign Mgr	Regional Mgr	Store Mgr	Store Staff	Method	Priority
User Invited	Email	Email	Email	Email	Email	Email	Email	HIGH
User Activated	In-App	In-App	In-App	In-App	In-App	In-App	In-App	LOW
User Deactivated	Email	Email	Email	Email	Email	Email	Email	HIGH
Password Reset Requested	Email	Email	Email	Email	Email	Email	Email	CRITICAL
Role Changed	-	Email	Email	Email	Email	Email	Email	NORMAL
Access Revoked	Email	Email	Email	Email	Email	Email	Email	HIGH

B.8 Report & Export Notifications

Event	PSP Admin	Brand Admin	Campaign Mgr	Regional Mgr	Store Mgr	Store Staff	Method	Priority
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Scheduled Report Ready	Email	Email	Email	Email	-	-	Email	NORMAL
Export Complete	Email	Email	Email	Email	Email	-	Email	NORMAL
Export Failed	Email	Email	Email	Email	-	-	Email	HIGH
Dashboard Threshold Alert	Push	Push	Push	Push	-	-	Push	HIGH

B.9 Escalation Notifications

Event	PSP Admin	Brand Admin	Campaign Mgr	Regional Mgr	Store Mgr	Store Staff	Method	Priority
Store Overdue (No Activity)	-	Email	Email	Push, Email	Push, Email	Push	Multi	CRITICAL
Retake Overdue (48h)	-	Push	Push	Push, Email	Push, Email	Push	Multi	CRITICAL
Campaign Compliance Risk	-	Push, Email	Push, Email	Push	-	-	Multi	CRITICAL
SLA Breach Warning	Push, Email	Push, Email	Push	-	-	-	Multi	CRITICAL

B.10 Deinstall Notifications

Event	PSP Admin	Brand Admin	Campaign Mgr	Regional Mgr	Store Mgr	Store Staff	Method	Priority
Deinstall Task Created	-	In-App	In-App	In-App	Push	Push	Multi	HIGH
Deinstall Reminder (3d before)	-	-	-	-	Push	Push	Push	NORMAL
Deinstall Due Today	-	In-App	In-App	In-App	Push	Push	Multi	HIGH

Deinstall Overdue	-	Email	Email	Push, Email	Push, Email	Push	Multi	CRITICAL
Deinstall Completed	-	In-App	In-App	In-App	In-App	-	In-App	NORMAL

B.11 Notification Preferences

B.11.1 User-Configurable Preferences

Users can customize notification delivery for non-critical events:

Preference	Options	Default
Email Frequency	Immediate, Daily Digest, Weekly	Immediate
Push Notifications	On, Off	On
Quiet Hours	Time range (e.g., 10pm-7am)	None
Email Format	HTML, Plain Text	HTML

B.11.2 Non-Configurable (Mandatory) Notifications

The following notification types cannot be disabled:

- Password reset
- Account deactivation
- Critical escalations (SLA breach)
- Security alerts
- System maintenance

B.12 Notification Templates

B.12.1 Email Subject Line Patterns

Event Category	Subject Pattern
Campaign	[NewPOPSys] {Campaign Name}: {Action}
Store	[NewPOPSys] Store {Store Name}: {Action Required}
Fulfillment	[NewPOPSys] Order #{OrderID}: {Status Update}
Issue	[NewPOPSys] Issue #{IssueID}: {Status}
User	[NewPOPSys] Account: {Action}
Escalation	[URGENT] NewPOPSys: {Escalation Type}

B.12.2 Push Notification Character Limits

Platform	Title Limit	Body Limit
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iOS	50 chars	150 chars
Android	65 chars	240 chars
Web Push	50 chars	120 chars

B.13 Webhook Events

For integration users, the following events trigger webhooks:

Event	Payload	Retry Policy
campaign.published	Campaign object	3 retries, exponential backoff
campaign.completed	Campaign object + summary	3 retries
order.created	Order + lines	3 retries
order.shipped	Order + shipment + tracking	3 retries
order.delivered	Order + delivery confirmation	3 retries
issue.created	Issue object	3 retries
issue.resolved	Issue object + resolution	3 retries
store.completed	StoreAssignment + rollup	3 retries
photo.submitted	Photo metadata (no image)	3 retries
photo.reviewed	Photo + decision	3 retries

B.14 Notification Delivery SLAs

Priority	Delivery Target	Retry Attempts
CRITICAL	< 30 seconds	5
HIGH	< 2 minutes	3
NORMAL	< 10 minutes	3
LOW	< 1 hour	2

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Appendix C: Export Formats

C.1 Overview

This appendix defines the specifications for all data exports supported by the NewPOPSys platform. Exports enable offline analysis, integration with external systems, and archival of campaign data.

Supported Export Formats

Format	Extension	Use Case	Max Records
CSV	.csv	Data analysis, spreadsheet import	100,000
Excel	.xlsx	Formatted reports with multiple sheets	50,000
PDF	.pdf	Printable reports, compliance documentation	N/A
JSON	.json	API integration, data migration	100,000

Export Access by Role

Role	Campaign Data	Store Data	Order Data	Photo Data	Financial
Platform Admin	Full	Full	Full	Full	Full
PSP Admin	Full	Full	Full	Metadata	Full
Brand Admin	Own Brands	Own Brands	Own Brands	Full	Limited
Campaign Manager	Assigned	Assigned	Assigned	Full	No
Regional Manager	Region Only	Region Only	Region Only	Region Only	No
Store Manager	Own Store	Own Store	Own Store	Own Store	No

C.2 CSV Export Specifications

C.2.1 General CSV Format

Property	Specification
Encoding	UTF-8 with BOM
Delimiter	Comma (,)
Quote Character	Double quote (")
Escape Character	Double quote ("")
Line Ending	CRLF (\r\n)
Header Row	Required (first row)
Date Format	ISO 8601 (YYYY-MM-DD)
DateTime Format	ISO 8601 (YYYY-MM-DDTHH:mm:ssZ)
Boolean Format	true / false
Null Values	Empty string

C.2.2 Campaign Export (CSV)

Filename Pattern: campaigns_{tenant}_{YYYYMMDD}.csv

Column	Type	Description
campaign_id	ULID	Unique campaign identifier
campaign_name	String	Campaign display name
brand_name	String	Associated brand name
campaign_type	Enum	PROMOTIONAL / CORE_BRANDING
status	Enum	Current campaign status
start_date	Date	Campaign start date
end_date	Date	Campaign end date (nullable)
store_count	Integer	Total assigned stores
completed_count	Integer	Stores completed
completion_rate	Decimal	Percentage complete (0.00-100.00)
created_at	DateTime	Creation timestamp
created_by	String	Creator email
published_at	DateTime	Publication timestamp (nullable)

C.2.3 Store Assignment Export (CSV)

Filename Pattern: store_assignments_{campaign_id}_{YYYYMMDD}.csv

Column	Type	Description
assignment_id	ULID	Unique assignment identifier
campaign_id	ULID	Parent campaign ID
campaign_name	String	Campaign display name
store_id	ULID	Store identifier
store_number	String	Store number/code
store_name	String	Store display name
region	String	Region name
district	String	District name (nullable)
territory	String	Territory name (nullable)
status	Enum	Assignment status
phase	Enum	Computed store phase

items_total	Integer	Total kit items
items_received	Integer	Items received
items_installed	Integer	Items installed
items_satisfied	Integer	Items approved
photos_submitted	Integer	Photos uploaded
photos_approved	Integer	Photos approved
photos_pending	Integer	Photos awaiting review
photos_rejected	Integer	Photos rejected
issues_open	Integer	Open issue count
assigned_at	DateTime	Assignment timestamp
completed_at	DateTime	Completion timestamp (nullable)

C.2.4 Order Export (CSV)

Filename Pattern: `orders_{campaign_id}_{YYYYMMDD}.csv`

Column	Type	Description
order_id	ULID	Order identifier
order_number	String	Human-readable order number
campaign_id	ULID	Campaign identifier
store_id	ULID	Store identifier
store_number	String	Store number
store_name	String	Store name
status	Enum	Order status
item_count	Integer	Total line items
total_quantity	Integer	Total units ordered
shipped_quantity	Integer	Units shipped
delivered_quantity	Integer	Units delivered
created_at	DateTime	Order creation timestamp
confirmed_at	DateTime	PSP confirmation (nullable)
shipped_at	DateTime	First shipment (nullable)
delivered_at	DateTime	Delivery confirmed (nullable)

tracking_numbers	String	Comma-separated tracking numbers
carrier	String	Carrier name

C.2.5 Issue Export (CSV)

Filename Pattern: `issues_{campaign_id}_{YYYYMMDD}.csv`

Column	Type	Description
issue_id	ULID	Issue identifier
issue_number	String	Human-readable issue number
campaign_id	ULID	Campaign identifier
store_id	ULID	Store identifier
store_number	String	Store number
issue_type	Enum	MISSING / DAMAGED / INCORRECT / PACKAGING
status	Enum	Issue status
item_sku	String	Affected item SKU
item_name	String	Affected item name
quantity_affected	Integer	Units affected
reported_by	String	Reporter email
reported_at	DateTime	Report timestamp
resolution	String	Resolution notes (nullable)
resolved_at	DateTime	Resolution timestamp (nullable)
reorder_id	ULID	Linked reorder (nullable)

C.3 Excel Export Specifications

C.3.1 General Excel Format

Property	Specification
Format	Office Open XML (.xlsx)
Excel Version	Compatible with Excel 2010+
Max Rows per Sheet	50,000
Date Format	Localized (user preference)
Number Format	Localized (user preference)

Header Row	Frozen, bold, filtered
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C.3.2 Campaign Summary Report (XLSX)

Filename Pattern: campaign_report_{campaign_name}_{YYYYMMDD}.xlsx

Sheet 1: Summary

Row	Content
1-5	Report header (title, date range, generated by)
7	Campaign details (name, brand, dates)
9-15	KPI summary (stores, completion %, issues, etc.)

Sheet 2: Store Details

Full store assignment data with conditional formatting:

- Green: Complete stores
- Yellow: In progress
- Red: Overdue/Exception

Sheet 3: Item Breakdown

Kit item summary with quantities:

- Ordered vs Shipped vs Delivered vs Installed

Sheet 4: Issues

All issues with status and resolution

Sheet 5: Timeline

Daily progress chart data

C.3.3 Compliance Report (XLSX)

Filename Pattern: compliance_report_{campaign_name}_{YYYYMMDD}.xlsx

Sheet 1: Executive Summary

- Overall compliance rate
- Regional breakdown
- Risk highlights

Sheet 2: Store Compliance

Column	Description
Store	Store name and number
Region	Geographic region
Items Required	Total required items

Items Satisfied	Approved items
Compliance %	Satisfaction rate
Status	COMPLIANT / NON-COMPLIANT / WAIVED
Outstanding Items	Items still pending
Days Since Start	Days in campaign

Sheet 3: Non-Compliant Stores

Filtered list of non-compliant stores with:

- Missing items detail
- Pending retakes
- Open issues
- Last activity date

Sheet 4: Waiver Log

All waived items with justification

C.4 PDF Report Specifications

C.4.1 General PDF Format

Property	Specification
Page Size	Letter (8.5" x 11") or A4
Orientation	Portrait (default) or Landscape
Margins	0.75" all sides
Font	Sans-serif (Helvetica/Arial)
Logo	Tenant logo in header
Page Numbers	Bottom right
Generated Date	Bottom left

C.4.2 Campaign Completion Certificate (PDF)

Purpose: Official documentation of successful campaign completion at a store.

Template Structure:

+-----+	
[TENANT LOGO]	Campaign Completion
	Certificate
+-----+	
This certifies that	

	[STORE NAME]	
	Store #[STORE_NUMBER]	
	[STORE ADDRESS]	
	has successfully completed all requirements	
	for the campaign:	
	[CAMPAIGN NAME]	
	[BRAND NAME]	
	Completion Date: [DATE]	
	Items Installed: [COUNT]	
	Photos Approved: [COUNT]	
	Verified by: [REVIEWER NAME]	
	Certificate ID: [ULID]	
+-----+		
	Page 1 of 1 Generated: [TIMESTAMP]	
+-----+		

C.4.3 Photo Proof Report (PDF)

Purpose: Visual documentation of installed materials with approval status.

Template Structure:

- Page 1: Cover page with campaign and store info
- Pages 2-N: Photo grid (2x2 per page)
 - Photo thumbnail (300x300px)
 - Item name
 - Location slot
 - Status (Approved/Rejected)
 - Reviewer notes
 - Timestamp

C.4.4 Executive Summary Report (PDF)

Purpose: High-level campaign performance for stakeholders.

Sections:

1. **Cover Page**
 - Campaign name, brand, date range
 - Report generated date
2. **KPI Dashboard** (1 page)
 - Total stores / Complete / In Progress
 - Completion rate gauge chart
 - Issue rate
 - Photo approval rate
3. **Regional Breakdown** (1 page)

- Bar chart by region
- Top/bottom performers

4. **Timeline** (1 page)

- Cumulative completion curve
- Daily submission rates

5. **Issues Summary** (1 page)

- Issue counts by type
- Resolution rates

6. **Appendix** (as needed)

- Detailed store list

C.5 JSON Export Specifications

C.5.1 General JSON Format

Property	Specification
Encoding	UTF-8
Format	Pretty-printed (2-space indent)
Date Format	ISO 8601
Null Handling	Explicit null
Empty Arrays	[]
Root Element	Object with metadata + data array

C.5.2 Standard JSON Wrapper

```
{
  "export": {
    "type": "campaign_data",
    "version": "1.0",
    "generated_at": "2026-01-01T12:00:00Z",
    "generated_by": "user@example.com",
    "tenant_id": "01ABC...",
    "filters": {
      "campaign_id": "01DEF...",
      "date_from": "2026-01-01",
      "date_to": "2026-01-31"
    },
    "record_count": 150,
    "page": 1,
    "total_pages": 1
  },
  "data": [
```



```
{ ... },
{ ... }
]
}
```

C.5.3 Campaign Object (JSON)

```
{
  "id": "01HXYZ...",
  "name": "Summer 2026 Promo",
  "brand": {
    "id": "01HABC...",
    "name": "Brand Co"
  },
  "type": "PROMOTIONAL",
  "status": "PUBLISHED",
  "dates": {
    "start": "2026-06-01",
    "end": "2026-08-31"
  },
  "kit": {
    "id": "01HKIT...",
    "item_count": 5,
    "items": [
      {
        "id": "01HITM...",
        "sku": "WC-24x36-001",
        "name": "Window Cling 24x36",
        "quantity": 2
      }
    ]
  },
  "rollup": {
    "stores_total": 500,
    "stores_complete": 425,
    "stores_in_progress": 50,
    "stores_not_started": 25,
    "completion_rate": 85.0
  },
  "metadata": {
    "created_at": "2026-05-01T10:00:00Z",
    "created_by": "admin@example.com",
    "published_at": "2026-05-15T09:00:00Z"
  }
}
```

C.5.4 Store Assignment Object (JSON)

```

{
  "id": "01HASG...",
  "campaign_id": "01HXYZ...",
  "store": {
    "id": "01HSTR...",
    "number": "1234",
    "name": "Downtown Location",
    "address": {
      "street": "123 Main St",
      "city": "Anytown",
      "state": "CA",
      "postal_code": "90210",
      "country": "US"
    },
  },
  "geography": {
    "region": "West",
    "district": "SoCal",
    "territory": "LA Metro"
  },
},
"status": "COMPLETE",
"phase": "COMPLETE",
"items": {
  "total": 5,
  "received": 5,
  "installed": 5,
  "satisfied": 5
},
"photos": {
  "required": 5,
  "submitted": 6,
  "approved": 5,
  "rejected": 1
},
"issues": {
  "total": 1,
  "open": 0,
  "resolved": 1
},
"timeline": {
  "assigned_at": "2026-06-01T00:00:00Z",
  "ready_at": "2026-06-05T14:30:00Z",
  "started_at": "2026-06-06T10:00:00Z",
  "submitted_at": "2026-06-06T16:00:00Z",
  "completed_at": "2026-06-07T09:30:00Z"
}
}

```

C.6 Data Export Schemas

C.6.1 Available Export Endpoints

Endpoint	Description	Formats
/exports/campaigns	Campaign list with rollups	CSV, XLSX, JSON
/exports/campaigns/{id}/stores	Store assignments for campaign	CSV, XLSX, JSON
/exports/campaigns/{id}/orders	Orders for campaign	CSV, XLSX, JSON
/exports/campaigns/{id}/issues	Issues for campaign	CSV, XLSX, JSON
/exports/campaigns/{id}/photos	Photo metadata (not images)	CSV, JSON
/exports/campaigns/{id}/report	Formatted report	XLSX, PDF
/exports/stores/{id}/history	Store campaign history	CSV, JSON

C.6.2 Export Request Parameters

Parameter	Type	Description
format	Enum	csv, xlsx, pdf, json
date_from	Date	Filter start date
date_to	Date	Filter end date
status	Enum[]	Filter by status(es)
region	String[]	Filter by region(s)
include_photos	Boolean	Include photo URLs (JSON only)
page	Integer	Page number (pagination)
per_page	Integer	Records per page (max 10000)

C.6.3 Export Response

```
{
  "export_id": "01HEXP...",
  "status": "PROCESSING",
  "progress": 45,
  "estimated_completion": "2026-01-01T12:05:00Z",
  "download_url": null,
  "expires_at": null
}
```

When complete:

```
{
  "export_id": "01HEXP...",
  "status": "COMPLETE",
}
```

```
"progress": 100,  
"completed_at": "2026-01-01T12:03:00Z",  
"download_url": "https://cdn.example.com/exports/01HEXP....csv",  
"expires_at": "2026-01-02T12:03:00Z",  
"file_size_bytes": 1048576,  
"record_count": 5000  
}
```

C.7 Photo Export Specifications

C.7.1 Photo Metadata Export (CSV/JSON)

Photos are exported as metadata with signed URLs (not embedded images).

Field	Description
photo_id	Unique photo identifier
assignment_id	Parent assignment
item_id	Associated kit item
location_slot	Location slot name
status	PENDING / APPROVED / REJECTED
original_url	Signed URL (expires 24h)
thumbnail_url	Signed thumbnail URL
uploaded_at	Upload timestamp
reviewed_at	Review timestamp
reviewer	Reviewer email
rejection_reason	Reason code (if rejected)

C.7.2 Photo Archive Export (ZIP)

For bulk photo download, a ZIP archive is generated:

Structure:

```
photos_{campaign_id}_{YYYYMMDD}.zip  
├─ manifest.json  
├─ store_1234/  
│   ├─ item_ABC_slot_1.jpg  
│   ├─ item_ABC_slot_1_retake.jpg  
│   └─ item_DEF_slot_2.jpg  
├─ store_5678/  
│   └─ ...  
└─ rejected/  
    └─ ...
```

Manifest.json:

```
{
  "campaign_id": "01HXYZ...",
  "export_date": "2026-01-01",
  "total_photos": 2500,
  "total_size_mb": 1250,
  "stores": [
    {
      "store_number": "1234",
      "photo_count": 5,
      "files": [
        {
          "filename": "item_ABC_slot_1.jpg",
          "item_sku": "WC-24x36-001",
          "status": "APPROVED"
        }
      ]
    }
  ]
}
```

C.8 Scheduled Exports

C.8.1 Schedule Configuration

Field	Description
frequency	DAILY / WEEKLY / MONTHLY
day_of_week	For weekly (0-6)
day_of_month	For monthly (1-28)
time	UTC time (HH:MM)
format	Export format
recipients	Email addresses
filters	Standard filter criteria

C.8.2 Delivery Options

Option	Description
Email Attachment	Attached to email (max 10MB)
Email Link	Download link in email
SFTP	Upload to configured SFTP server
Webhook	POST to configured URL

S3 Bucket	Upload to customer S3 bucket
-----------	------------------------------

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Appendix D: Glossary

D.1 Abbreviations and Acronyms

Abbreviation	Full Term	Description
API	Application Programming Interface	Standard protocol for software component communication
AWS	Amazon Web Services	Cloud computing platform used for hosting and storage
BullMQ	Bull Message Queue	Queue library for Node.js based on Redis for background jobs
CI/CD	Continuous Integration / Continuous Deployment	Automated software delivery pipeline
CSV	Comma-Separated Values	Plain text data format for tabular data
DDL	Data Definition Language	SQL statements for schema definition
E2E	End-to-End	Testing methodology covering full user flows
ERD	Entity Relationship Diagram	Database schema visualization
HMAC	Hash-based Message Authentication Code	Cryptographic authentication mechanism for webhooks
JSON	JavaScript Object Notation	Lightweight data interchange format
JSONB	JSON Binary	PostgreSQL binary JSON data type with indexing
JTBD	Jobs To Be Done	Product development methodology focusing on user goals
LTS	Long-Term Support	Software version with extended maintenance
MIS	Management Information System	Business reporting and analytics system
NFR	Non-Functional Requirement	System quality attributes (performance, security, etc.)
ORM	Object-Relational Mapping	Database abstraction layer (Drizzle)
OTLP	OpenTelemetry Protocol	Standard for transmitting telemetry data
PK	Primary Key	Unique database record identifier

POP	Point of Purchase	Marketing materials displayed at retail locations
PSP	Print Service Provider	Company producing and fulfilling POP materials
PWA	Progressive Web App	Web application with native app capabilities
RBAC	Role-Based Access Control	Permission system based on user roles
REST	Representational State Transfer	API architectural style
S3	Simple Storage Service	AWS object storage for files and photos
SDK	Software Development Kit	Tools for building applications
SKU	Stock Keeping Unit	Unique product identifier
SLA	Service Level Agreement	Performance and availability guarantees
SOW	Statement of Work	Project scope and requirements document
SQL	Structured Query Language	Database query language
SRS	Software Requirements Specification	This document
SSO	Single Sign-On	Unified authentication across systems
SUPP	Supplemental Specification	Additional technical requirements document
TBD	To Be Determined	Placeholder for undefined requirements
UI	User Interface	Visual elements users interact with
ULID	Universally Unique Lexicographically Sortable Identifier	Time-ordered unique ID format
UUID	Universally Unique Identifier	Standard unique ID format
UX	User Experience	Overall user satisfaction and usability
XLSX	Excel Spreadsheet Format	Microsoft Excel file format
XML	Extensible Markup Language	Data interchange format

D.2 Domain Terminology

D.2.1 Core Business Entities

Term	Definition
PspTenant	Root tenant representing a PSP organization. The paying customer. All brands, users, and data belong to a tenant. Multi-tenant isolation is enforced at the database level.
Brand	A client brand managed by the PSP. Each brand has its own stores, campaigns, and settings. Represents the end customer of the PSP.

Store	A physical retail location where POP materials are installed. Identified by store number within a brand. Has geographic hierarchy assignment.
Campaign	A brand's promotional program with defined timeline, participating stores, and materials to install. The primary unit of work in the system.
Kit	The collection of POP items to be shipped and installed for a campaign. Defined by Brand Admin during campaign setup.
KitItem	A single item within a kit (e.g., "Window Cling 24x36", "Counter Card"). Has SKU, quantity, and photo requirements.

D.2.2 Campaign Types

Term	Definition
Promotional Campaign	Campaign with a defined end date that triggers deinstall workflow upon expiration. Materials are temporary. Also called "Expiring Campaign".
Core Branding Campaign	Campaign with no end date. Deinstall only triggered manually by admin action. Materials are permanent branding elements. Also called "Non-expiring Campaign" or "Evergreen Campaign".

D.2.3 Geographic Hierarchy

Term	Definition
Region	Top level of geographic hierarchy (required). Examples: "West Coast", "Northeast", "Central". Used for Regional Manager scoping.
District	Second level of geographic hierarchy (optional). Groups stores within a region. Examples: "SoCal", "Bay Area".
Territory	Third level of geographic hierarchy (optional). Most granular geographic grouping. Examples: "LA Metro", "SF Downtown".
StoreGroup	Custom, non-geographic grouping of stores. Used for targeting campaigns to specific store sets. A store can belong to multiple groups. Examples: "Premium Stores", "24-Hour Locations".

D.2.4 Survey and Layout

Term	Definition
SurveyTemplate	A reusable survey definition created by Brand Admin. Contains questions, sections, conditional logic, and validation rules.
SurveyVersion	An immutable snapshot of a SurveyTemplate. Campaigns pin to a specific version to ensure consistency across stores.
StoreLayout	Definition of a store's physical advertising locations. Describes where POP items can be placed. Versioned to support changes over time.
LocationSlot	A specific placement within a store where POP items can be installed. Examples: "Front Window - Left", "Counter Display - Register 1".

PhotoRule	Requirements for proof photos at a location slot. Specifies minimum/maximum photos, resolution requirements, and capture instructions.
------------------	--

D.2.5 Fulfillment Terms

Term	Definition
StoreOrder	An order generated when a campaign is published, representing what PSP needs to produce and ship to a store. One order per store per campaign.
OrderLine	A line item in a StoreOrder specifying SKU, quantity required, and fulfillment status. Maps to KitItems.
Shipment	A physical shipment sent by PSP to a store. An order may have multiple shipments (partial fulfillment). Contains carrier and tracking information.
ShipmentLine	Items included in a specific shipment. Enables tracking partial fulfillment when not all items ship together.
Reorder	A separate order generated to replace missing or damaged items reported through the issue workflow.
Batch	PSP grouping of orders for production efficiency. Types: PRODUCTION (manufacturing), PICK_PACK (warehouse), SHIP_WAVE (logistics), CUSTOM (ad-hoc).

D.2.6 Execution and Verification

Term	Definition
StoreAssignment	The record binding a Store to a Campaign. Tracks progress through the complete lifecycle from assignment to completion.
AssignmentItem	Individual kit item tracked at the store level. Maintains quantities and status through fulfillment and execution.
ReceiveVerification	Store's confirmation that a shipment was received with item-level acceptance, damage reporting, and quantity verification.
CompletionSubmission	Store's submission indicating installation is complete, including attestation checkbox and final comments.
PhotoUpload	A proof photo uploaded by the store. Tied to an item and/or location slot. Includes metadata like timestamp and GPS.
PhotoReview	An admin's review decision on a photo. States: PENDING, APPROVED, REJECTED. Includes reason codes for rejections.
RetakeRequest	A request for the store to retake a rejected photo. Includes reviewer feedback and deadline.

D.2.7 Issue Management

Term	Definition
------	------------

IssueRequest	A store-reported problem requiring resolution. Created during receipt verification or later in the process.
IssueLine	A specific item and quantity affected by an issue. Multiple lines can be attached to one IssueRequest.
IssueType	Categories: MISSING (item not in shipment), DAMAGED (item arrived damaged), INCORRECT (wrong item shipped), PACKAGING (packaging issue only).
IssueResolution	How the issue was resolved: REORDER (replacement shipped), CREDIT (refund issued), WAIVED (no action needed), DENIED (not approved).

D.2.8 Workflow Terms

Term	Definition
Campaign Publish	Action that transitions campaign from DRAFT to PUBLISHED. Generates store assignments and orders. Triggers notifications to stores.
Receipt Survey	First-stage store survey upon delivery. Confirms items received, reports issues, captures receipt photos if required.
Install Survey	Second-stage store survey. Captures proof photos of installed items at designated location slots.
Completion Attestation	Store user's checkbox confirmation that installation is complete. Required before submission.
Retake Loop	Workflow cycle: rejected photos trigger RetakeRequest, store uploads new photo, admin reviews, repeat until approved or waived.
Reorder Flow	Issue approved triggers Reorder generation, PSP produces and ships replacement, store receives and verifies.
Deinstall Task	End-of-campaign task where store removes expired POP materials. Created automatically when promotional campaign ends.
Waiver	Administrative override marking a requirement satisfied without proof. Used for exceptions and edge cases. Requires justification.
Reopen	Admin action to unlock a COMPLETE store assignment for additional work. Creates audit trail.
Split Shipment	Order fulfilled via multiple shipments. Common when items have different production times.

D.2.9 Technical Terms

Term	Definition
Rollup	Aggregation of lower-level quantities to higher levels. Item quantities roll up to Store, Store metrics roll up to Campaign.

Pinning	Freezing a specific version of survey or layout to a store assignment. Ensures consistency even if template changes.
Rebase	Updating a pinned version to latest. May require re-verification. Admin action with confirmation.
Idempotency Key	Unique identifier ensuring API request processed only once. Prevents duplicate orders or actions on retry.
Soft Delete	Marking records as deleted (deleted_at timestamp) without physical removal. Preserves audit history.
Event Outbox	Pattern for reliable webhook/event dispatch with retries. Ensures external systems receive all notifications.
Tenant Isolation	Database-level separation ensuring one tenant cannot access another's data. Enforced via Row-Level Security (RLS).

D.2.10 Verification Modes

Term	Definition
STRICT Mode	Every photo requires explicit approval by reviewer before store can complete. Default for high-compliance campaigns.
FAST Mode	Photos auto-approve unless explicitly flagged. Store completes immediately on submission. Used for trusted stores or low-risk campaigns.
Hybrid Mode	Some items require strict verification, others auto-approve. Configured per KitItem.

D.3 User Roles (Personas)

D.3.1 PSP Level Roles

Role	Definition
Platform Admin	Highest privilege. Full system configuration, tenant management, user impersonation for support. Internal PSP IT staff.
PSP Admin	Brand onboarding, PSP-level settings, user management, reporting and exports. Day-to-day PSP operations manager.
Production Operator	Updates order statuses, creates shipments and tracking, processes batches. Warehouse and fulfillment staff.

D.3.2 Brand Level Roles

Role	Definition
Brand Admin	Full brand configuration, all campaigns access, store management, user permissions. Primary brand contact at PSP.

Campaign Manager	Brand Admin scoped to assigned campaigns only. Manages specific campaigns without full brand access.
Regional Manager	Exception-first workflow. Reviews photos, handles escalations, scoped to assigned region. Field operations role.

D.3.3 Store Level Roles

Role	Definition
Store Manager	Store-level authority. Manages team, approves replacements, full execution permissions. Store general manager.
Store Operator	Execution-only permissions. Complete surveys, update status, request replacements. Store associate or merchandiser.

D.3.4 System Level Roles

Role	Definition
Integration User	Service account for API/webhook operations. No UI access. Used for system-to-system communication.

D.4 Status Values

D.4.1 Campaign Status

Status	Description
DRAFT	Being configured, not active
SCHEDULED	Set to publish at future date
PUBLISHED	Active, stores executing
COMPLETED	All stores satisfied or ended
CANCELLED	Terminated before completion
ARCHIVED	Historical, read-only

D.4.2 Store Assignment Status

Status	Description
ASSIGNED	Selected for campaign
READY	Materials delivered
IN_PROGRESS	Actively working
SUBMITTED	Photos pending review
COMPLETE	All requirements met

REWORK_REQUIRED	Photos rejected
REOPENED	Unlocked for more work
WAIVED	Admin bypassed review

D.4.3 Photo Review Status

Status	Description
PENDING	Awaiting review
APPROVED	Accepted
REJECTED	Needs retake
SUPERSEDED	Replaced by retake

D.4.4 Issue Status

Status	Description
OPEN	Just reported
TRIAGED	Classified and routed
AWAITING_APPROVAL	Needs brand approval
APPROVED	Reorder authorized
DENIED	Rejected
IN_FULFILLMENT	Replacement shipping
RESOLVED	Replacement received

D.5 Industry Terms

Term	Definition
Point of Purchase (POP)	Marketing materials displayed at retail locations to influence buying decisions.
Point of Sale (POS)	Location where transactions occur; often used interchangeably with POP in retail context.
Planogram	Diagram showing how products and displays should be arranged in a store.
Merchandising	Activities to promote product sales including display setup and maintenance.
Compliance	Degree to which stores meet campaign requirements for installation and photo proof.
Field Execution	On-site activities at retail locations to implement marketing programs.

Fulfillment	Process of producing, packaging, and shipping materials to stores.
Kitting	Assembling multiple items into a single package for shipping.
Proof of Performance	Documentation (typically photos) proving marketing materials were properly installed.
Retail Audit	Verification visit to confirm in-store execution meets standards.

Document Version: 1.0 Last Updated: 2026-01-01 Source: SRS 1.3 Definitions, SUPP-001, Domain Expert Interviews

Appendix E: Change Log

E.1 Overview

This appendix maintains a chronological record of all changes to the Software Requirements Specification (SRS) document. All modifications, additions, and deletions are tracked with version numbers, dates, authors, and descriptions.

Version Numbering Convention

Version Type	Format	Description
Major	X.0	Significant structural changes, new modules, or breaking changes
Minor	X.Y	New features, sections, or substantial additions
Patch	X.Y.Z	Corrections, clarifications, or minor updates

E.2 Change History

Version	Date	Author	Section(s) Affected	Description
1.0	2026-01-01	SRS Team	All	Initial SRS document creation. Complete specification including: Introduction, Overall Description, System Architecture, User Personas & RBAC, Module Specifications (Shared Foundations, Mobile PWA, Brand Admin, PSP Operations, Store Portal), and Appendices.

E.3 Detailed Change Records

Version 1.0 - Initial Release (2026-01-01)

Summary: Complete initial Software Requirements Specification for NewPOPSys platform.

Sections Delivered

Section	Description	Status
---------	-------------	--------

00_Meta	Document metadata, progress tracking, source mapping	Complete
01_Introduction	Purpose, scope, definitions, references, overview	Complete
02_Overall_Description	Product perspective, functions, user classes, environment, constraints	Complete
03_System_Architecture	Database model, application architecture, technology stack, integrations	Complete
04_User_Personas_RBAC	Persona matrix, permissions, authentication flows, 9 persona profiles	Complete
05_Module_SharedFoundations	Universal login screen specification	Complete
06_Module_MobilePWA	Mobile/PWA screens (Login, Dashboard, Receipt Survey, Install Survey)	Partial
07_Module_BrandAdmin	Brand admin screens (Dashboard, Campaign List, Store Selection)	Partial
08_Module_PSPOperations	PSP operations screens (Order Queue, Shipments, Issues)	Complete
09_Module_StorePortal	Store portal screens (Dashboard, History, Gallery, Team, Reports)	Complete
99_Appendices	State machines, notifications, export formats, glossary, change log	Complete

Key Features Specified

1. Multi-Tenant Architecture

- PSP tenant isolation with Row-Level Security
- Brand hierarchy with geographic organization
- Store-level execution tracking

2. Campaign Management

- Full campaign lifecycle (Draft → Published → Completed → Archived)
- Kit definition with photo requirements
- Store assignment and targeting

3. Fulfillment Workflow

- Order generation and tracking
- Shipment management with carrier integration
- Issue reporting and reorder processing

4. Execution & Verification

- Receipt verification workflow
- Installation photo capture
- Review and approval process
- Retake request handling

5. Integration Architecture

- RESTful API design
- Webhook event system
- External carrier integrations
- S3 file storage

E.4 Pending Changes

ID	Requested Date	Description	Priority	Target Version
PC-001	TBD	Complete remaining Mobile PWA screens (M005-M008)	High	1.1
PC-002	TBD	Complete remaining Brand Admin screens (B004-B007)	High	1.1
PC-003	TBD	Add Platform Admin module specification	Medium	1.1
PC-004	TBD	Add API Specifications section (11.x)	High	1.1
PC-005	TBD	Add Non-Functional Requirements section (12.x)	High	1.1

E.5 Review and Approval

Version 1.0 Approval

Role	Name	Date	Signature
Project Sponsor	_____	//__	_____
Product Owner	_____	//__	_____
Technical Lead	_____	//__	_____
QA Lead	_____	//__	_____

E.6 Change Request Template

For future change requests, use the following template:

```

### Change Request: [CR-XXX]

**Date Submitted**: YYYY-MM-DD
**Submitted By**: [Name]
**Priority**: [Critical | High | Medium | Low]

#### Description
[Detailed description of the proposed change]
```


Justification

[Business or technical reason for the change]

Sections Affected

- [] Section X.X - Description
- [] Section Y.Y - Description

Impact Assessment

- ****Scope Impact****: [None | Minor | Moderate | Major]
- ****Schedule Impact****: [None | Minor | Moderate | Major]
- ****Resource Impact****: [None | Minor | Moderate | Major]

Approval

- [] Product Owner Approval
- [] Technical Lead Approval
- [] Stakeholder Notification

Implementation Notes

[Any technical notes for implementing the change]

E.7 Document Governance

Review Schedule

Review Type	Frequency	Participants
Quarterly Review	Every 3 months	Product Owner, Tech Lead, QA Lead
Major Release Review	Per release	Full project team
Ad-hoc Review	As needed	Relevant stakeholders

Change Control Process

1. **Request**: Submit change request using template
2. **Review**: Technical lead reviews for feasibility
3. **Approve**: Product owner approves priority and scope
4. **Implement**: Author updates relevant sections
5. **Verify**: QA reviews changes for consistency
6. **Publish**: Updated version released with change log entry